

Endogenous Cap-and-trade NYISO Model With Zonal Trading

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Motivation

New York State's Climate Act has currently set ambitious emissions targets including a 40% reduction in emissions from 1990's levels by 2030 (Climate Act, 2024). In order to achieve these goals, the government must enact policies that will accelerate the retirement of emitting energy sources and the expansion of clean energy. Currently, there are two major mechanisms for the expansion of renewables: a carbon tax, which implements a tax per tonne of carbon emitted, and a cap and trade model, which sets an overall carbon cap and allows for companies to buy and sell permits to emit CO₂. While the carbon tax is often considered a simple and effective approach, cap and trade can be a more economically viable approach that allows for gradual renewable energy adoption while ensuring carbon emissions targets are still met. The cap and trade model often involves the setting of an initial carbon emissions cap that can be lowered each subsequent year, allowing companies to prepare to adjust their portfolios over time. The initial carbon limit set and the rates of decline can have important implications for the success of the policy in reducing emissions and its market impact.

The model presented in this report is intended to simulate a cap-and-trade system within the two major regions of New York: an upstate region (zones A-F) and a downstate region (zones G-K). Within the upstate region, renewable resources have larger existing capacities, and higher availability, while accessibility within the downstate region is less available. The goal of the project is to investigate how state policy can accelerate the decarbonization of electricity, and how cap-and-trade can allow for a more feasible and economically viable transition into a

renewable portfolio given the contrasting resources of these two regions. The model will analyze a 10 year period spanning 2025-2034 with 2024 data serving as an initial condition for installed capacity. Inputs to the model will consist of low, moderate, and high initial caps (28 TW, 31 MW, 33 MW) and graduate, moderate, and steep rates of decrease (4%, 6%, 8%), for a total of 9 solutions. Each solution will be examined to determine the relative amount of installed renewable capacity in each zone and when this capacity is installed, cumulative emissions, and the carbon price associated with each scenario.

Cap and Trade Historical Context

The model implementation is motivated by the study of various real-world examples of cap-and-trade systems in Europe and the United States. The general form of a cap-and-trade system involves government regulation of pollutant emissions in a specific industry or group of industries. For the purposes of this project, the regulated pollutant will be carbon dioxide. In cap-and-trade systems, a government or regulatory agency will set an initial cap on emissions and reduce this cap over time, often by a linear factor each year. Governments then distribute allowances, or emissions permits, to companies through auctions or for free. This distribution occurs at set time intervals, often quarterly. If companies emit less carbon dioxide than their purchased allowances, they can bank these allowances for future time periods or trade them with other companies. If companies emit more carbon dioxide than their purchased allowances, they must pay a penalty (EDF, 2020).

Within this standard cap-and-trade framework, there are many implementation decisions that impact the cost and rate of decarbonization for each company as well as the efficiency of decarbonizing the industry as a whole. The table below shows several decision factors involved

in designing and implementing a cap-and-trade system as well as hypothesized impacts from loosening and restricting these factors (Kenton, 2025). This report will focus on varying the yearly reduction factor, but other parameters may be considered in future stages of analysis.

There are more factors besides those in the table that have been implemented in real systems but will not be considered in this model, such as setting allowance price floors and gradually reducing the number of free allowances over time (CMUA, 2021).

Table 1

Impacts of Cap and Trade Decision Variables on Policy Outcome

Decision Factor	Too High	Too Low
Total emissions cap	No meaningful emissions reduction	Carbon leakage, misreporting, high allowance prices
Yearly reduction factor	Carbon transition too aggressive, unsustainable rate of carbon price increase	Carbon transition too slow, lack of incentive for early investment in renewables
Over-emission penalty cost	High allowance price ceiling	Low compliance and cap becomes non-binding
Share of permits allocated with no cost	Companies may over-emit to increase baseline allowances	Carbon leakage, high cost of compliance

Literature Review

To inform our methodology we researched examples of cap and trade modeling in academic papers. We researched papers doing different types of cap and trade and also utilizing different model parameters such as retirement, transmission and granularity to help us build a model that asks the question we want to be asking. The first paper was about the different types of cap and trade(Chen, Chen, & Ma, 2021). The novel thing they uncovered was in the costs and

benefits of Grandfathering (GM) Vs Benchmarking(BM). Under BM, a unit carbon quota is set for a firm by the government to restrict unit carbon emissions. That is, a firm is allowed to emit a certain number of units of carbon per unit of energy produced. Under GM, a total carbon emission quota is set for a firm for a certain period, based on that firm's previous production levels to restrict the total carbon emissions. Both of these policies have been implemented in real situations but they target different things. With GM electricity price increases directly with carbon price, utility firms will set the highest electricity prices, however firms will overall generate lower carbon emissions compared to BM. For BM prices are a secondary effect of carbon quotas, and despite not decreasing carbon emissions as much as GM, BM does push for significantly more new renewable energy capacity to be installed due to higher profitability. The second paper formulates a minimum-cost Generation Expansion Planning (GEP) model as a standard Linear Program(Mejía Giraldo, López Lezama, & Gallego Pareja, 2012). It's distinctive in combining three environmental levers simultaneously, a carbon tax, an annual emissions cap with a fixed reduction rate, and mandatory retirement of aging plants. The transmission network is also included, which matters because it allows optimal capacity to be sited away from load centers. This paper found that using direct carbon tax was variably effective. With no emissions policy, coal persists at 26% through year 20, emissions double. Adding \$0.01/lb tax + 5%/yr reduction rate results in coal eliminated by year 7, replaced by NGCC/nuclear/wind. The "Green" system costs 40% more but achieves 93% emissions reduction vs. baseline.

Methods

Formulation

1 Sets and Parameters

Sets:

$t \in \mathcal{T} = \{1, \dots, 10\}$	years
$z \in \mathcal{Z} = \{1, 2\}$	zones (AF, GK)
$g \in \mathcal{G}$	generation technologies
$h \in \mathcal{H}$	sampled hours
$k \in \mathcal{W}$	representative weeks

Representative weeks are selected using k -means clustering with scale factor $\sigma = 8760/|\mathcal{H}|$.

Financial parameters:

$$r = 0.07, \quad \delta_t = (1 + r)^{-(t-1)}, \quad n = 30$$

$$\text{CRF} = \frac{r(1 + r)^n}{(1 + r)^n - 1}, \quad p = \$400/\text{tCO}_2$$

Cost parameters:

$c_{g,t}^{\text{inv}}$: overnight capex (\$/MW)

$c_{g,t}^{\text{fom}}$: fixed O&M (\$/MW-yr)

$c_{g,t}^{\text{op}}$: variable O&M (\$/MWh)

Battery costs: $c_t^{b,\text{inv}}, c_t^{b,\text{fom}}$

Demand and availability:

$D_{z,h}$: hourly demand

$\lambda_{z,t}$: load growth scalar

$\bar{D}_{z,t}$: annual peak demand

$a_{g,z,h} \in [0, 1]$: generator availability

Other parameters:

Parameter	Value	Parameter	Value
Gas emissions	0.439	Gas capacity credit	0.92
Coal emissions	0.935	Coal capacity credit	0.85
Nuclear emissions	0.012	Nuclear capacity credit	0.98
Hydro emissions	0.060	Hydro capacity credit	0.90
Wind emissions	0.113	Wind capacity credit	0.30
Solar emissions	0.352	Solar capacity credit	0.06
Battery credit κ^b	0.50	Duration Δ	4 hr
Efficiency η	0.85	Transfer limit $\bar{F}$	2000 MW
Initial cap $\bar{C}_1$	33×10^6	Decline rate γ	0.08

Carbon cap:

$$\bar{C}_t = \max(\bar{C}_1 - \gamma \bar{C}_1(t-1), 5 \times 10^6)$$

Build limits (MW/yr):

Gas AF: 500, Gas GK: 4500, Wind: 1500, Solar: 1000, Battery: 1000

Retirement limits (MW/yr):

Gas: 800, Nuclear: 1200, Hydro: 200, Wind: 500, Solar: 300, Battery: 500

2 Decision Variables

$x_{g,z,t}^+, x_{g,z,t}^-$	capacity additions/retirements
$X_{g,z,t}$	installed generation capacity
$y_{g,z,t,h}$	dispatch
$\tau_{t,h}$	AF→GK transfer
$E_{z,t,h}$	hourly emissions
$A_{z,t}, B_{z,t}, M_{z,t}$	allowances, bank, penalty purchases
$x_{z,t}^{b+}, x_{z,t}^{b-}, X_{z,t}^b$	battery capacity variables
$b_{z,t,h}^c, b_{z,t,h}^d, s_{z,t,h}$	charge, discharge, SOC

All variables are nonnegative unless otherwise noted.

3 Model Formulation

Capacity Expressions

$$X_{g,z,t} = x_{g,z}^0 + \sum_{\tau=1}^t (x_{g,z,\tau}^+ - x_{g,z,\tau}^-) \quad (\text{E1})$$

$$X_{z,t}^b = X_z^{b,0} + \sum_{\tau=1}^t (x_{z,\tau}^{b+} - x_{z,\tau}^{b-}) \quad (\text{E2})$$

$$\hat{E}_{z,t} = \sum_{h \in \mathcal{H}} E_{z,t,h} \quad (\text{E3})$$

Objective Function

$$\begin{aligned} \min \quad & \sum_{t,g,z} \delta_t c_{g,t}^{\text{inv}} x_{g,z,t}^+ + \sum_{t,g,z} \delta_t c_{g,t}^{\text{fom}} X_{g,z,t} \\ & + \sigma \sum_{t,g,z,h} \delta_t c_{g,t}^{\text{op}} y_{g,z,t,h} + \sum_{t,z} \delta_t c_t^{b,\text{inv}} x_{z,t}^{b+} \\ & + \sum_{t,z} \delta_t c_t^{b,\text{fom}} X_{z,t}^b + \sum_{t,z} \delta_t p M_{z,t} \end{aligned} \quad (\text{P})$$

Constraints

Power balance

$$\sum_g y_{g,1,t,h} + b_{1,t,h}^d - b_{1,t,h}^c - \tau_{t,h} = \lambda_{1,t} D_{1,h} \quad (\text{C1})$$

$$\sum_g y_{g,2,t,h} + b_{2,t,h}^d - b_{2,t,h}^c + \tau_{t,h} = \lambda_{2,t} D_{2,h} \quad (\text{C2})$$

Generation and transmission

$$y_{g,z,t,h} \leq a_{g,z,h} X_{g,z,t} \quad (\text{C3})$$

$$-\bar{F} \leq \tau_{t,h} \leq \bar{F} \quad (\text{C4})$$

Emissions and cap-and-trade

$$E_{z,t,h} = \sum_g e_g y_{g,z,t,h} \quad (\text{C5})$$

$$\sum_z A_{z,t} = \bar{C}_t \quad (\text{C6})$$

$$B_{z,t} = B_{z,t-1} + A_{z,t} - \sigma \hat{E}_{z,t} + M_{z,t} \quad (\text{C7})$$

with $B_{z,0} = 0$.

Battery operation

$$b_{z,t,h}^c, b_{z,t,h}^d \leq X_{z,t}^b \quad (C8)$$

$$s_{z,t,h} \leq \Delta X_{z,t}^b \quad (C9)$$

$$s_{z,t,h_k^{\text{first}}} = \eta b_{z,t,h_k^{\text{first}}}^c - b_{z,t,h_k^{\text{first}}}^d \quad (C10)$$

$$s_{z,t,h} = s_{z,t,h-1} + \eta b_{z,t,h}^c - b_{z,t,h}^d \quad (C11)$$

Resource adequacy

$$\sum_g \kappa_g X_{g,z,t} + \kappa^b X_{z,t}^b \geq \bar{D}_{z,t} \quad (C12)$$

$$X_{z,t}^b \geq \underline{X}_{z,t}^b \quad (C13)$$

Build and retirement limits

$$x_{g,z,t}^+ \leq \bar{x}_{g,z}^+ \quad (C14)$$

$$x_{g,z,t}^- \leq \bar{x}_{g,z}^- \quad (C15)$$

$$X_{g,z,t}, X_{z,t}^b \geq 0 \quad (C16)$$

Data Selection

Demand

Hourly load profiles for zones AF (NYISO Zones A–F) and GK (NYISO Zones G–K) are drawn from NYISO's aggregated hourly load records for the 2025 base year. Zone-specific annual growth scalars are derived from Table I-4a of the NYISO 2024 Gold Book, which reports forecast non-coincident summer peak demand by zone through 2034 (NYISO, 2024). Zone-level peaks are aggregated to AF and GK and normalized by the 2025 base year value, yielding dimensionless scalars that project AF load growing approximately 11% and GK load growing approximately 6.5% over the decade.

These scalars are applied in two places. The hourly dispatch constraint scales the base-year load profile by the year-specific scalar, preserving the shape of the profile while increasing its magnitude each year. The capacity adequacy constraint uses a peak demand equal

to the historical zone maximum multiplied by the same scalar, evaluated over the full 8,760-hour profile rather than the representative weeks sample to avoid underestimating the annual system peak.

Representative Weeks

To reduce computational burden, hourly dispatch is solved over a sample of 10 representative weeks (840 hours) rather than all 8,760 hours of the year. K-means clustering groups the 52 weeks of the year into 10 clusters of similar renewable availability profiles, and the week closest to each cluster centroid is selected as the representative. The algorithm is run with 30 random initializations and the solution minimizing mean absolute percentage error between full-year and sampled capacity factor means is retained.

A scale factor $\sigma = 8,760 / 840 \approx 10.43$ is applied wherever sampled sums are used to estimate annual totals. This occurs in the carbon bank balance constraint and the variable O&M cost term of the objective. Peak demand for the capacity adequacy constraint is evaluated over the full 8,760-hour profile to avoid underestimating the annual system peak.

Technology Cost

Capital and operating costs for all generating technologies are drawn from the NREL Annual Technology Baseline (ATB) 2024, Moderate scenario (NREL, 2024). The Moderate scenario represents NREL's central cost projection between the Conservative and Advanced cases. Three cost components are extracted for each technology across 2025–2034: overnight capital cost (\$/kW), fixed operations and maintenance cost (\$/kW-year), and variable operations and maintenance cost (\$/kWh). The technologies included are natural gas, coal, nuclear,

hydropower, wind, solar PV, and 4-hour utility-scale battery storage. Costs are processed from the raw ATB file using a Python script and converted from \$/kW to \$/MW for use in the model.

Because costs are time-indexed, declining renewable and storage costs directly influence investment timing. The optimizer can defer construction to capture lower future prices since capital costs are paid and discounted in the year of construction. Capital recovery factors use a 7% discount rate and a standard 30-year cost recovery period for all technologies, consistent with the ATB financial assumptions (NREL, 2024).

Capacity Credits

Capacity credits for conventional generators follow the NYISO Installed Capacity (ICAP) Manual methodology of $UCAP = ICAP \times (1 - EFORD)$, where EFORD is the Equivalent Demand Forced Outage Rate from NERC's Generating Availability Data System (NYISO, 2024a; NERC, 2024). For wind and solar, capacity credits reflect Effective Load Carrying Capability (ELCC) values consistent with NYISO's marginal capacity accreditation framework approved by FERC in 2022 (NYISO, 2024b). Hydro capacity credit is derived from the ratio of summer capability to nameplate capacity reported in the NYISO 2024 Gold Book Table III-3a. Battery storage capacity credit follows NYISO Installed Capacity Manual Section 5.12 accreditation.

Availability Factors

Availability factors for sources with effectively constant availability (gas, coal, hydropower, and nuclear) were assigned a singular availability factor based on Weighted Equivalent Forced-Outage (WEFO) values in the 2024 State of Reliability Overview provided by

the North American Electric Reliability Corporation (*2024 State of Reliability Overview*, 2024).

The report shows WEFO values of approximately 8% for gas, 15% for coal, 2% for nuclear, and 10% for hydro. These were subtracted from 100% to obtain availability percentages of 92%, 85%, 98%, and 90%, respectively. For wind and solar availability, data was aggregated from the ARPA-E PERFORM datasets published by the Open Energy Initiative ((Sergi et al., 2022)).

These datasets contained both solar and wind energy output every 5 minutes per zone, so individual data points were added together to create a dataset with hourly time intervals and total output within the upstate and downstate regions.

Existing Capacity

Initial installed capacity for each generator type and zone is drawn from NYISO 2024 Gold Book Table III-3a, which reports existing summer capability (MW) for all generating units participating in NYISO markets as of 2024. Summer capability is used rather than nameplate capacity because it reflects actual deliverable output under peak conditions and is consistent with how peak demand is measured in the capacity adequacy constraint.

Units are aggregated from NYISO's eleven individual zones to the two model zones and mapped to the six generator categories in the model based on fuel type. Battery storage initial capacity is drawn from Table III-2a using nameplate rather than summer capability, for consistency with how the CLCPA storage mandate counts installed MW. Most battery units show zero summer capability in the Gold Book, so nameplate capacity is considered more relevant for this model.

Emissions Factors

Generator CO₂ emission rates are derived from EPA eGRID 2023, which reports plant-level annual generation and emissions for all U.S. grid-connected facilities (EPA, 2024). A Python script computes generation-weighted average emission rates by technology class. Plant-level intensity is calculated as total plant CO₂ divided by fossil generator MWh only. Non-combustion generators (wind, solar, hydro, nuclear, battery) are assigned zero direct emissions regardless of co-location. Without this correction, solar and wind units co-located at fossil plants would incorrectly inherit the host plant's emission rate. The gas rate is a generation-weighted blend of combined cycle (0.430 tCO₂/MWh) and combustion turbine (0.524 tCO₂/MWh) units, yielding 0.439 tCO₂/MWh. Nuclear is not reported in eGRID and is assigned 0.012 tCO₂/MWh from IPCC lifecycle estimates (IPCC, 2022).

Policy Parameters

Two policy parameters are incorporated into the model, and these are grounded in New York State law. The carbon allowance cap follows a linear declining trajectory starting at a set amount (33, 31, or 28 MT) in 2025 and declining at a set rate (4%, 6%, 8%) per year. The penalty price for exceeding the cap was calibrated iteratively at \$400/tCO₂. This was found to be the threshold at which the model transitions from paying penalties to investing in sufficient clean capacity to remain within the cap.

The battery storage mandate is derived from CLCPA Section 66-p of the Public Service Law, which sets statutory targets of 3,000 MW statewide by 2030 and 6,000 MW by 2035 (NY PSL §66-p). In this model, annual intermediate values are linearly interpolated between these endpoints, as the statute specifies only the terminal targets. The statewide mandate is allocated to

the two model zones in proportion to their share of NYCA summer peak demand from the 2024 Gold Book Table I-4a: 35% to AF and 65% to GK.

Build and Retire Constraints

Build and retirement caps were retroactively added to the model formulation after several trials resulting in unrealistically rapid expansion and retirement of resources in year 1 and no changes in consequent years. In reality, large capacity additions and retirements are constrained by permitting and construction timelines among other factors, and should occur more gradually throughout the 10 year period. The specific limits are imposed based on the relative size and construction timeline for a plant of this type and are intended to prevent corner solutions rather than to precisely characterize these constraints.

Results

Capacity Evolution

In zones A-F, battery capacity closely corresponds with the minimum storage mandate in most scenarios, with the 82 MW present in year 1 reflecting existing capacity carried into the model from the 2024 baseline. This suggests storage investment in AF is driven primarily by the mandate rather than economic optimization. Solar capacity remains unchanged at 200 MW throughout, likely reflecting the relatively weak solar resource in New York and the model favoring wind as the lower-cost renewable option. Gas capacity declines over the first 4-5 years before stabilizing, with the 8% reduction scenario reaching a lower equilibrium than the 4% and 6% scenarios in most cases. Wind capacity behavior varies notably with the initial cap – under 31 MT and 33 MT caps, wind remains near 2,500 MW or rises modestly, while the 28 MT

scenario drives more substantial wind expansion across all reduction rates, indicating that only the tightest initial cap is sufficient to make additional wind investment economic in this region. Generally, the most pronounced capacity responses occur under the 8% reduction rate and 28 MT initial cap scenarios, consistent with these representing the most stringent policy conditions.

Figure 1

Zones A-F Cumulative Yearly Installed Capacity by Resource Type using 33 MT Initial Cap.

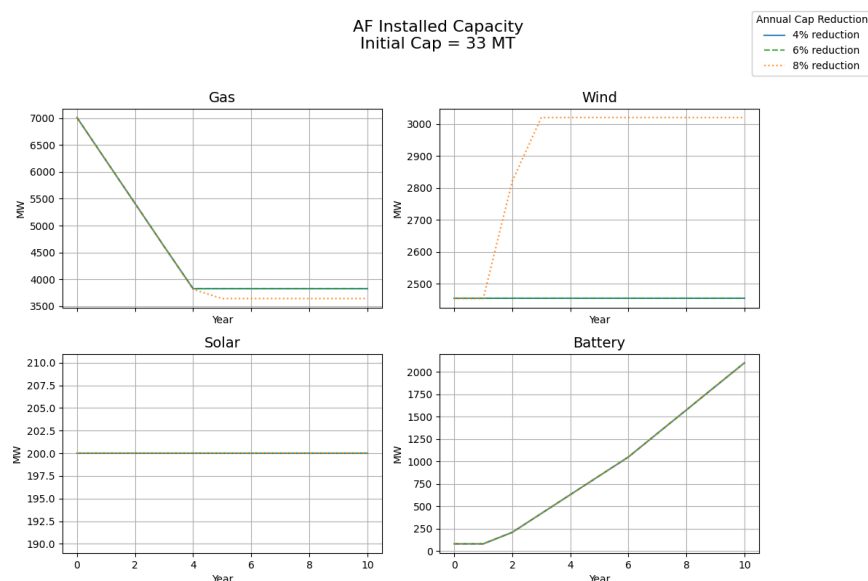


Figure 2

Zones A-F Cumulative Yearly Installed Capacity by Resource Type using 31 MT Initial Cap.

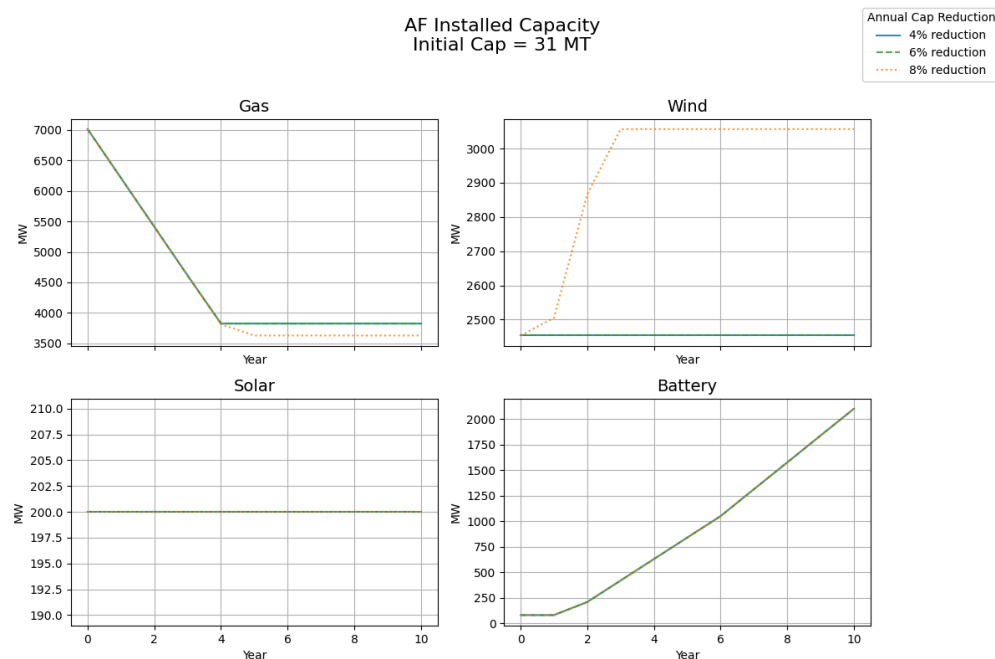
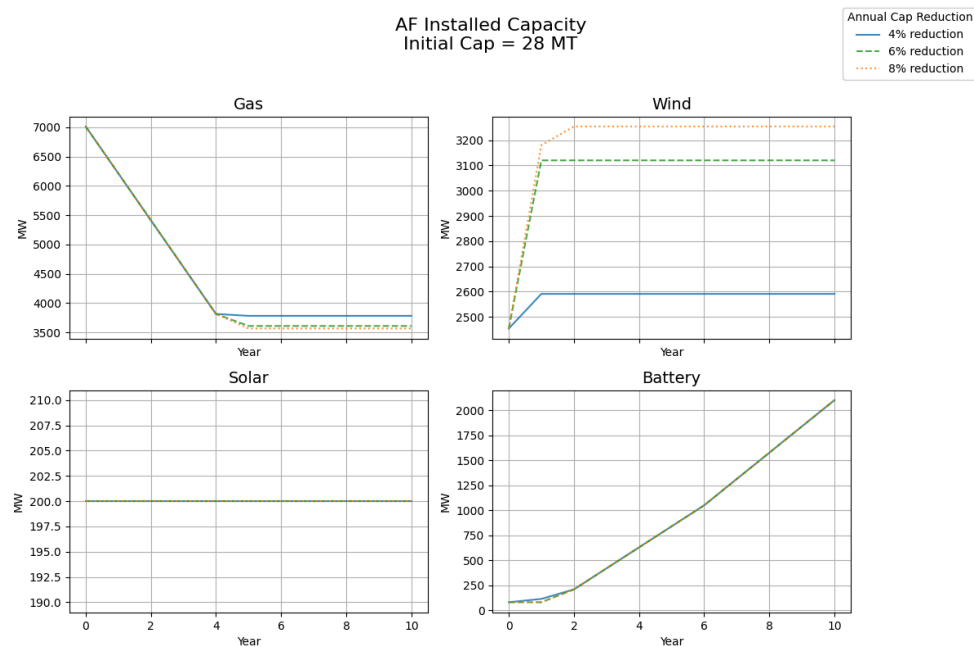


Figure 3

Zones A-F Cumulative Installed Capacity by Resource Type Using 28 MT Initial Cap.



Cumulative installed capacity in zones G-K displays several notable patterns. Gas capacity experiences a sharp increase of approximately 3,000 MW in year 1 across all three

initial cap scenarios, reflecting the relaxed build constraint discussed earlier, which was necessary for the model to reach a feasible solution. The model then immediately begins retiring gas capacity in subsequent years, suggesting that the year 1 expansion serves as a temporary buffer while renewable investment ramps up, rather than a long-run economic decision. Solar capacity remains flat under the 33 MT cap but begins expanding under the 31 MT and 28 MT caps, with the 28 MT and 8% reduction scenario reaching approximately 3,000 MW of final installed capacity. Wind capacity expands sharply across all scenarios, with the final settled capacity determined by the stringency of both the reduction rate and initial cap level. Battery capacity follows the mandated expansion path in later years but jumps immediately to 1,013 MW in the first three years across all nine scenarios, indicating that early storage investment is driven by the need to firm up renewable generation before sufficient wind and solar capacity is in place to effectively and cheaply meet demand.

Figure 4

Zones G-K Cumulative Yearly Installed Capacity by Resource Type using 33 MT Initial Cap.

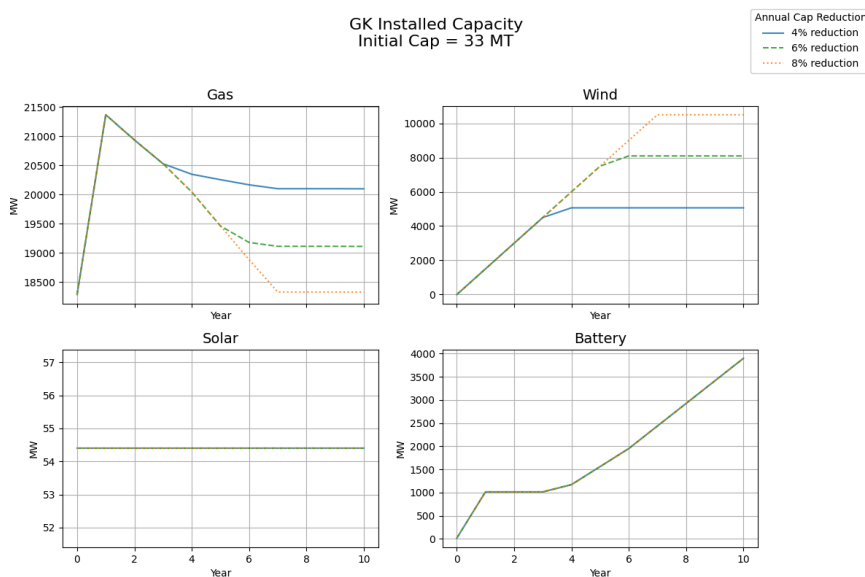
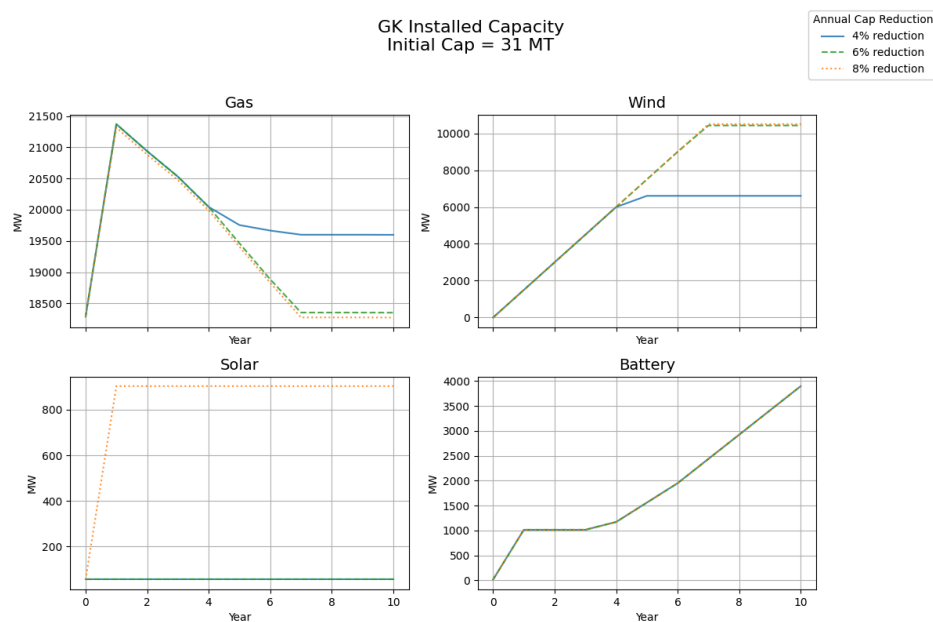
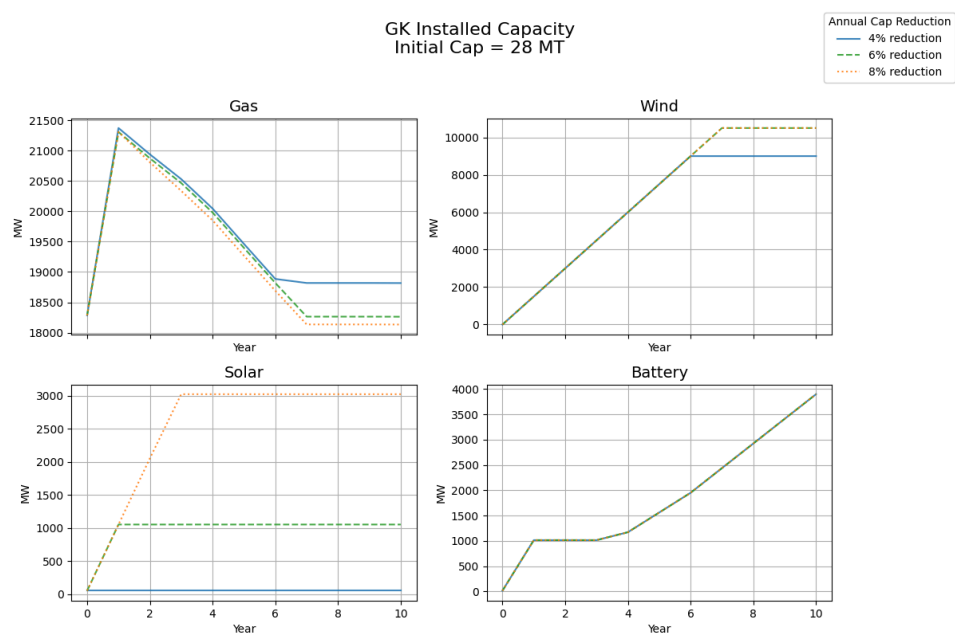


Figure 5

Zones G-K Cumulative Yearly Installed Capacity by Resource Type using 31 MT Initial Cap.

**Figure 6**

Zones G-K Cumulative Yearly Installed Capacity by Resource Type using 28MT Initial Cap.



Yearly Emissions

Yearly emissions vary greatly between the two regions. In the A-F region, emissions initially decrease slightly, and then sharply increase beginning in year 8. In the G-K region, emissions decrease steadily until they reach a point between 3-7 years at which they start slightly increasing. Within the G-K region, the model initially expands renewables and thus lowers its emissions until growth of demand starts outpacing growth of renewables, at which point emissions rise. The late-period emissions increase in zones A-F is consistent with the declining carbon shadow prices shown in Figure 9. As the shadow price falls, the implicit cost of emitting falls with it, reducing the incentive to further decarbonize. The cap becomes less binding at the margin, and the model increasingly dispatches existing gas capacity to meet demand rather than investing in additional abatement.

Figure 7

Zones A-F Yearly Emissions for Initial Caps of 28 MT, 31 MT, and 33 MT.

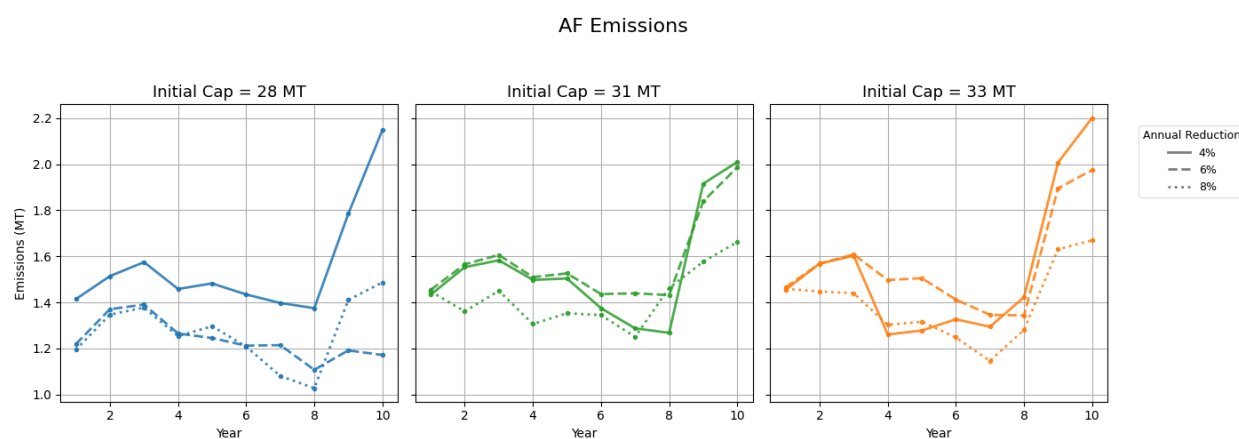
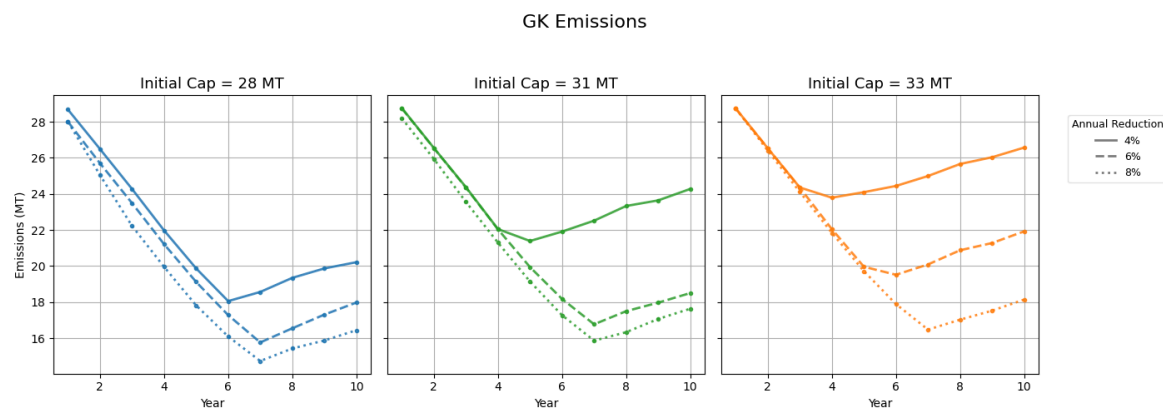


Figure 8

Zones G-K Yearly Emissions for Initial Caps of 28 MT, 31 MT, and 33 MT.

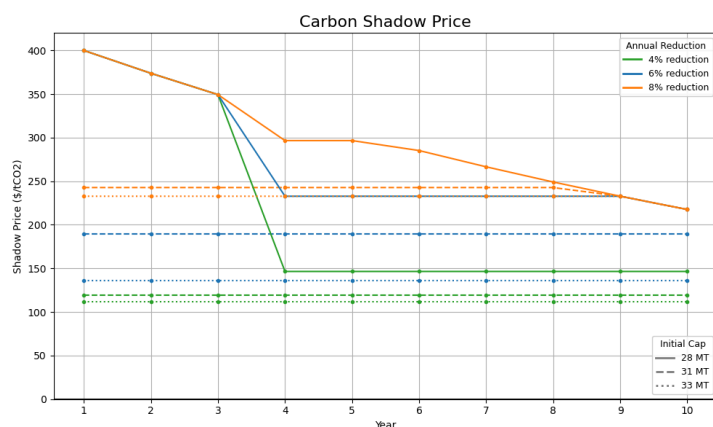


Yearly Carbon Price

A few trends emerge from Figure 9. Higher annual reduction rates correspond to higher carbon shadow prices, reflecting a more binding cap constraint. For the 28 MT initial cap, the shadow price begins at approximately \$400/tCO₂ and declines as early-period investment in clean capacity reduces the marginal cost of abatement. Under the 8% reduction scenario, the shadow price remains elevated and declines gradually throughout the horizon, indicating the cap continues to bind across all years. In contrast, the 4% and 6% scenarios see shadow prices stabilize at lower levels after year 4, suggesting the cap becomes less binding following the initial wave of investment.

Figure 9

Yearly Carbon Shadow Price for Varying Initial Carbon Caps and Annual Reduction Rates.



Conclusions/ Further Exploration

The results present several implications for the future of the NYISO capacity mix and the potential of cap and trade to support emissions goals. A central finding is that under realistic build constraints, continuously reducing annual emissions is not feasible within the modeled horizon without structural changes to the system. The model achieves early emissions reductions partly by relaxing the gas build constraint in zones G-K to maintain feasibility, effectively allowing the model to use gas as a short-run reliability buffer while renewable capacity is deployed. In later years, as renewable build limits are reached and transmission from zones A-F remains constrained, emissions begin to rise again. One potential remedy is increased out-of-state transmission – states with lower demand growth may be able to expand renewables faster than local demand increases, and regions with stronger solar resources could supplement New York's relatively weak solar generation.

The model also illustrates how cap and trade interacts differently across regions with varied resource mixes. In zones G-K, both the initial cap level and annual reduction rate meaningfully affect emissions trajectories, with lower initial caps having the strongest effect when reduction rates are modest, and the two parameters becoming less distinguishable as reduction rates increase. The results also suggest the model is somewhat more sensitive to changes in the reduction rate than to changes in the initial cap level, given that emissions differences across reduction rates appear larger than those across initial cap levels within the same scenario.

Overall, cap and trade can serve as an effective mechanism for reducing emissions in the NYISO system, but feasibility under more aggressive targets will likely require complementary

investments in transmission capacity, renewable build rates, or both. Future work could examine how relaxing these infrastructure constraints – through expanded interstate transmission, accelerated interconnection processing, or advances in renewable technology – affects the viability of more stringent cap trajectories.

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